

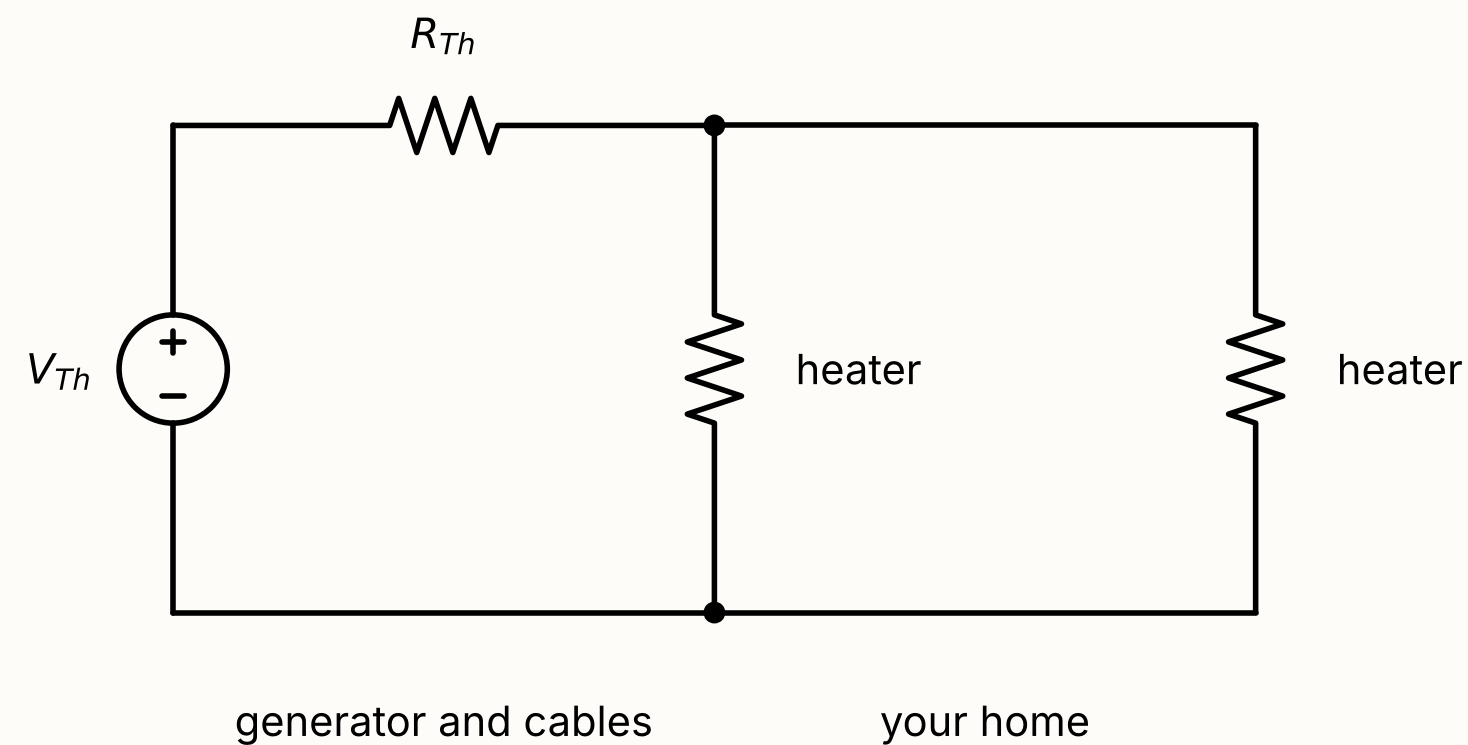
Thévenin Equivalents, Superposition and Maximum Power

Make a circuit simpler before you solve it, and answer what a load sees and how much power it can get.

Prepared by Dr. Adel Daaboul

PREDICT FIRST

The socket reads 230 V with nothing on and 200 V with one heater on. With two heaters?



about 170 V

about 177 V

about 185 V

One heater draws 10 A. The two heaters are identical.

TODAY

By the end of Part 2 you can

1 • Source transformations

Swap a voltage source and series resistor for a current source and parallel resistor, the arrow the right way.

2 • Superposition

Find a response one source at a time, and know why powers do not add.

3 • Thévenin and Norton

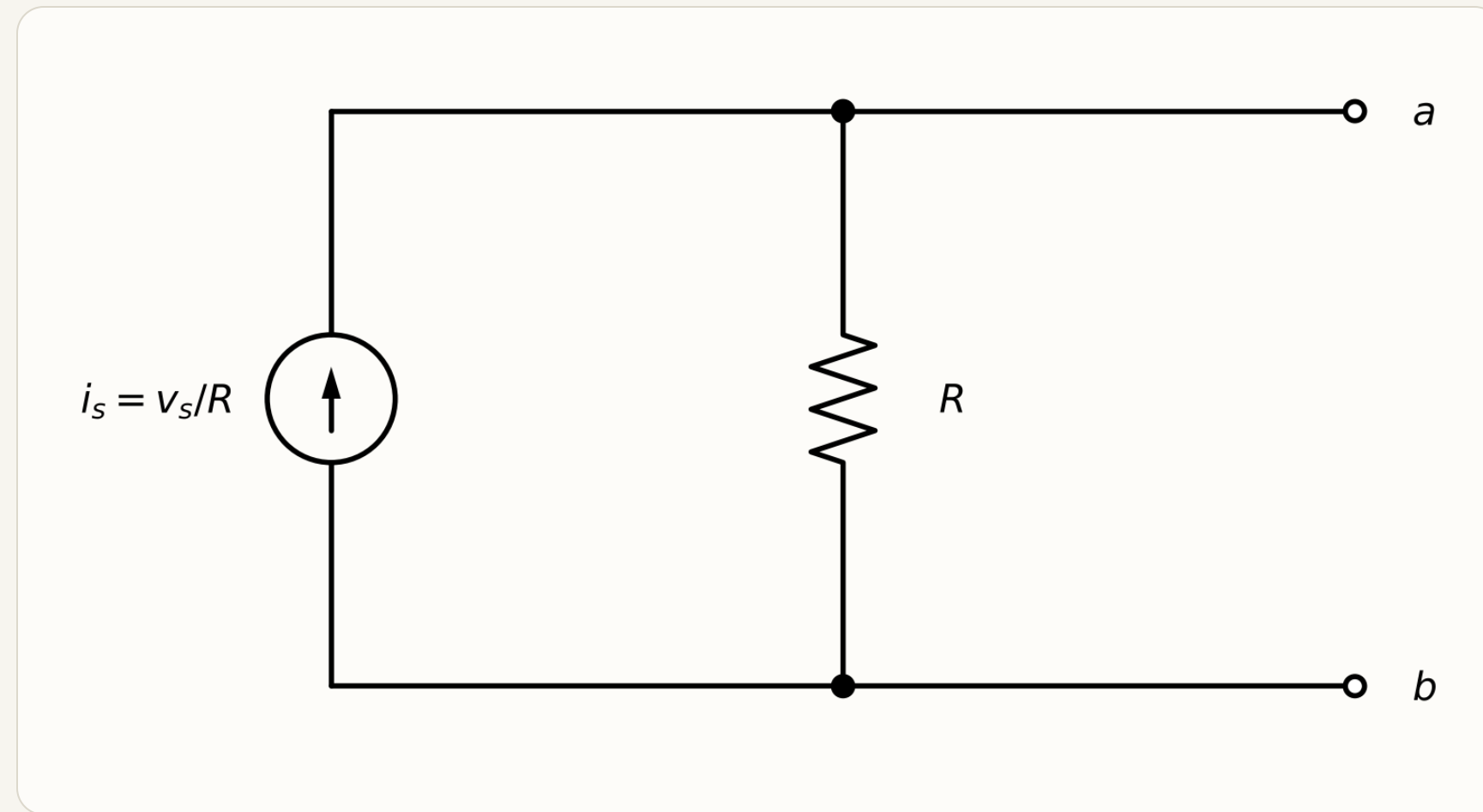
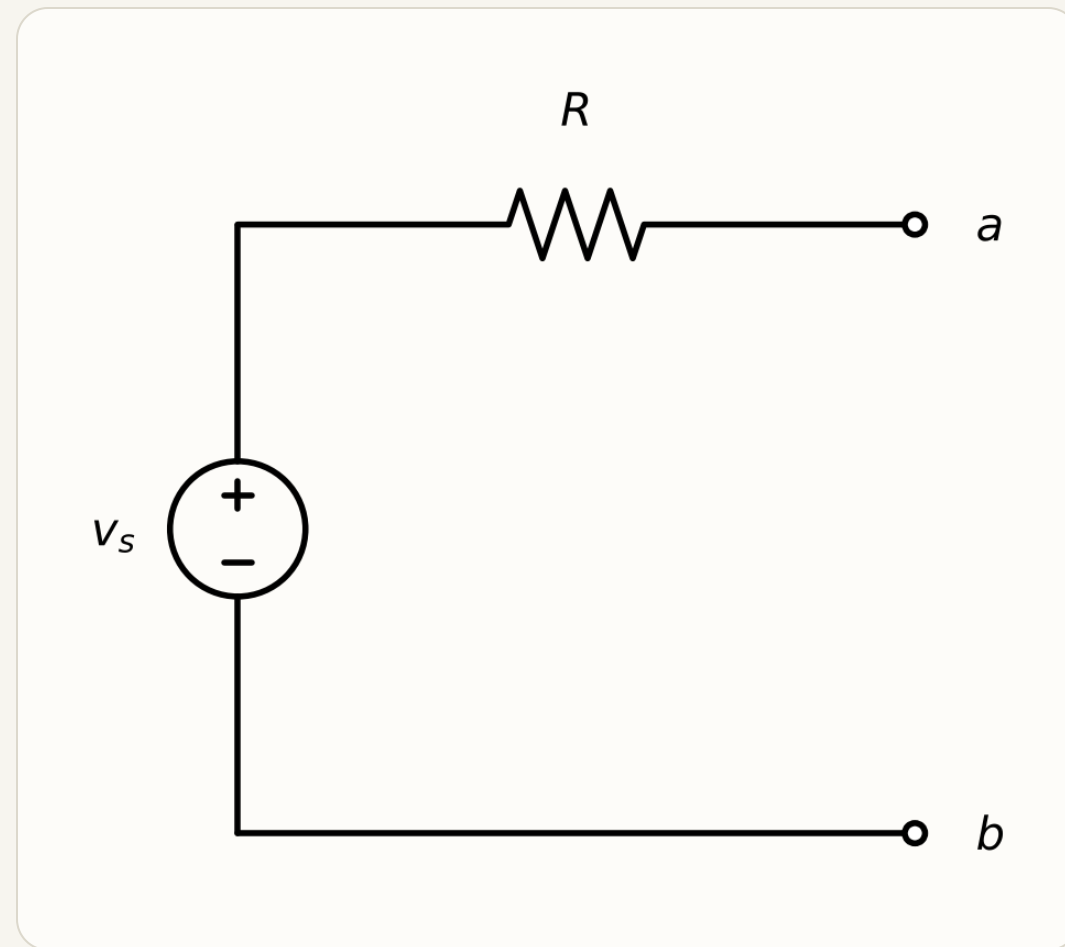
Find V_{Th} , i_{sc} and R_{Th} , with a test source when a dependent source is present.

4 • Maximum power

Choose R_L for maximum power and find that power and the efficiency.

SOURCE TRANSFORMATIONS

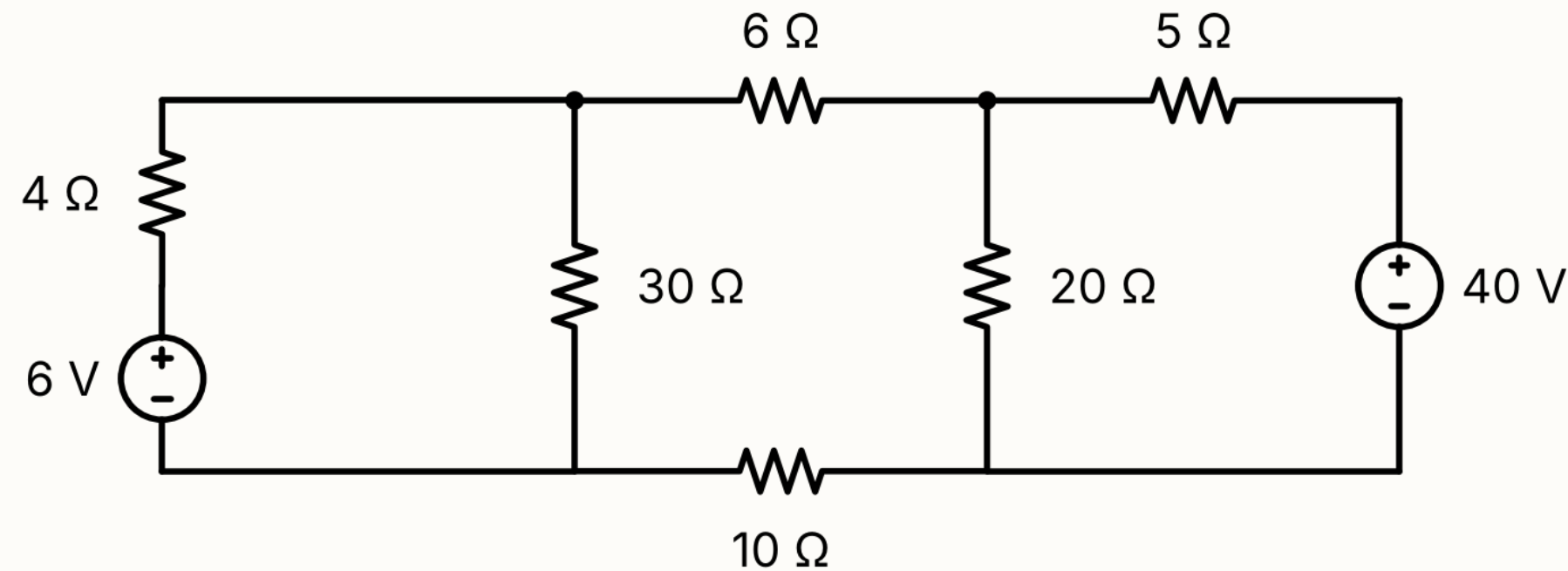
A series pair equals a parallel pair



$$i_s = \frac{v_s}{R}, \quad v_s = R i_s$$

The arrow points toward the terminal on the + side. The two circuits match only at a and b : the power inside them differs.

Transform everything except the branch you need



Find the power of the 6 V source. Start from the far end:

1. 40 V and 5 Ω $\rightarrow$ 8 A $\parallel$ 5 Ω
2. 20 $\parallel$ 5 = 4 Ω $\rightarrow$ 32 V with 4 Ω
3. 4 + 6 + 10 = 20 Ω $\rightarrow$ 1.6 A $\parallel$ 20 Ω
4. 20 $\parallel$ 30 = 12 Ω $\rightarrow$ 19.2 V with 12 Ω

NILSSON EXAMPLE 4.12

One loop is left: 19.2 V, 12 Ω , 4 Ω and the 6 V source

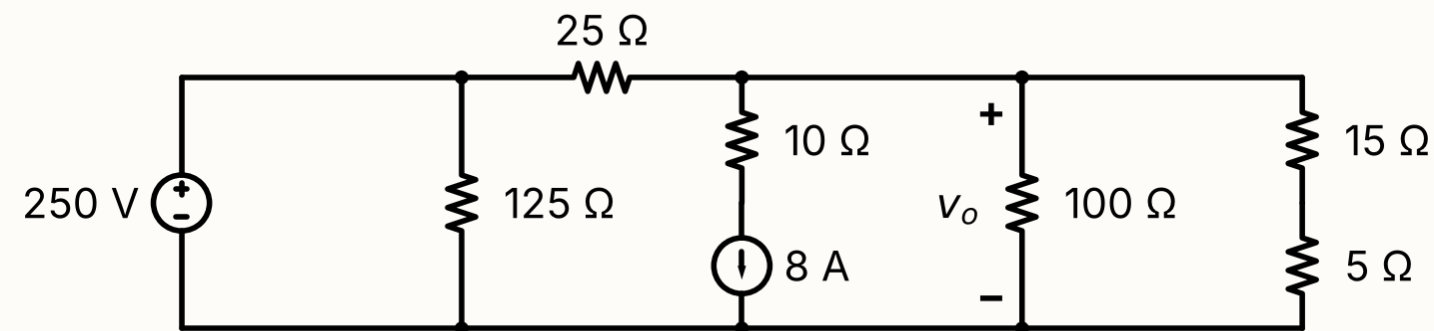
$$i = \frac{19.2 - 6}{12 + 4} = 0.825 \text{ A}, \quad p_{6 \text{ V}} = (6)(0.825) = 4.95 \text{ W}$$

The current enters the 6 V source's + terminal, so the source absorbs 4.95 W.

Check: solve the original circuit with node voltages. The 6 V branch carries the same 0.825 A.

NILSSON EXAMPLE 4.13

Find v_o first, then the powers



Drop the 125 Ω and the 10 Ω ; 250 V with 25 Ω becomes 10 A in parallel with 25 Ω :

$$\frac{250}{25} - 8 = 2 \text{ A}, \quad 25 \parallel 100 \parallel 20 = 10 \Omega \Rightarrow v_o = 20 \text{ V}$$

i_{250} : current out of the 250 V source's + terminal, in the original circuit:

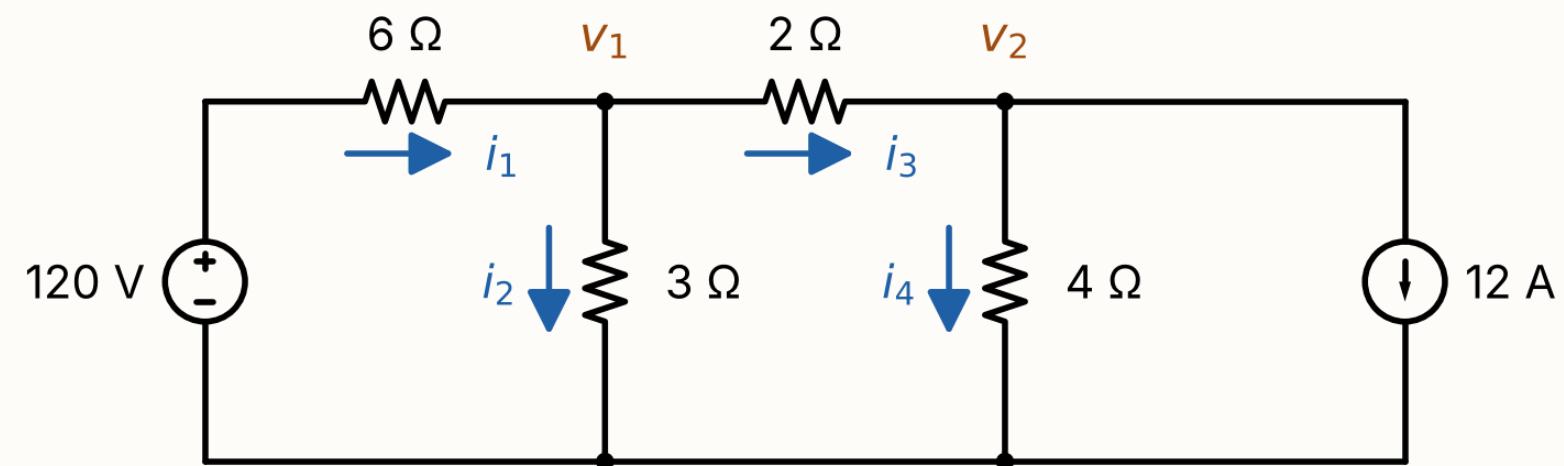
$$i_{250} = \frac{250}{125} + \frac{250 - 20}{25} = 11.2 \text{ A}, \quad p_{250 \text{ V}} = 2800 \text{ W developed}$$

$v_{8 \text{ A}}$: voltage across the 8 A source, + at the top:

$$v_{8 \text{ A}} = 20 - (8)(10) = -60 \text{ V}, \quad p_{8 \text{ A}} = 480 \text{ W developed}$$

SUPERPOSITION

Switch on one source at a time, then add the responses



Voltage source off

Replace it with a short circuit: 0 V.

Current source off

Replace it with an open circuit: 0 A.

Dependent sources stay

They are never switched off.

Nilsson Example 4.22: find i_1 to i_4 .

NILSSON EXAMPLE 4.22

Each source alone, then the sum

120 V alone (12 A source open):

$$\frac{v_1 - 120}{6} + \frac{v_1}{3} + \frac{v_1}{2 + 4} = 0 \Rightarrow v_1 = 30 \text{ V}$$

12 A alone (120 V source shorted):

$$\frac{v_1}{3} + \frac{v_1}{6} + \frac{v_1 - v_2}{2} = 0, \quad \frac{v_2 - v_1}{2} + \frac{v_2}{4} + 12 = 0$$

$$v_1 = -12 \text{ V}, \quad v_2 = -24 \text{ V}$$

	i ₁	i ₂	i ₃	i ₄
120 V alone	15 A	10 A	5 A	5 A
12 A alone	2 A	−4 A	6 A	−6 A
Both	17 A	6 A	11 A	−1 A

PREDICT FIRST

The currents add. Do the powers add too?

In Example 4.22 the $3\ \Omega$ carries 10 A with the 120 V source alone and -4 A with the 12 A source alone; together, 6 A.

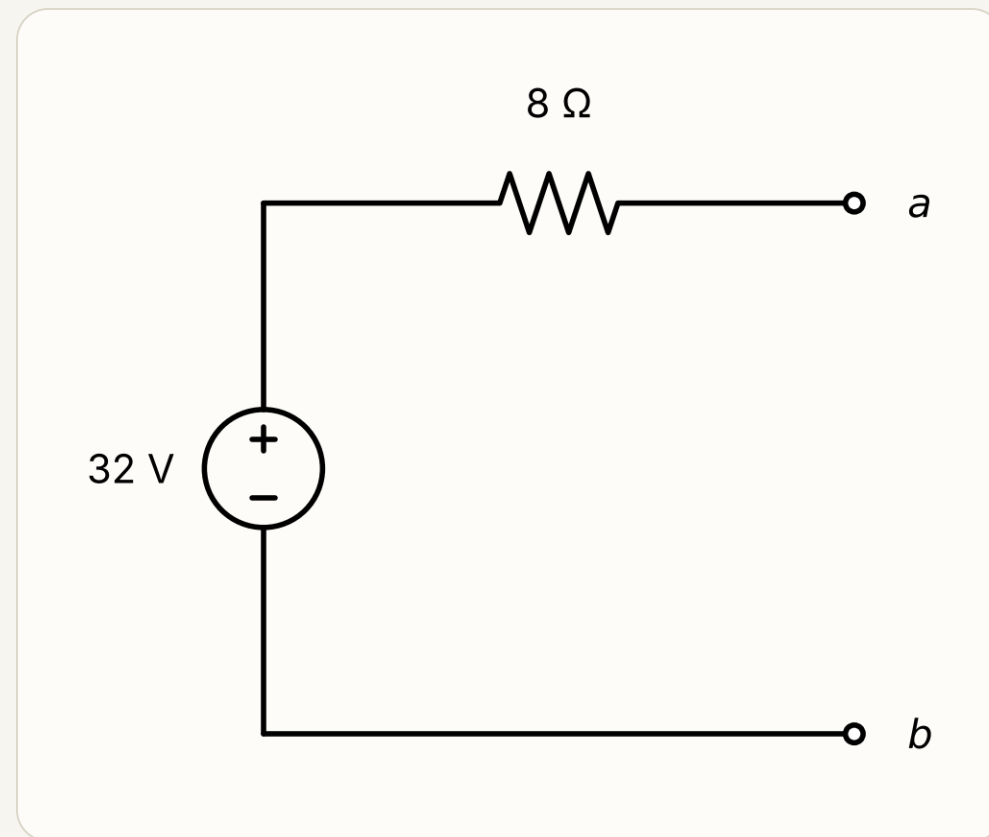
$$(10)^2(3) + (-4)^2(3) = 348\ \text{W} \neq (6)^2(3) = 108\ \text{W}$$

No. The circuit is linear, so currents and voltages add. Power goes as the square of the current, so add the currents first, then find the power.

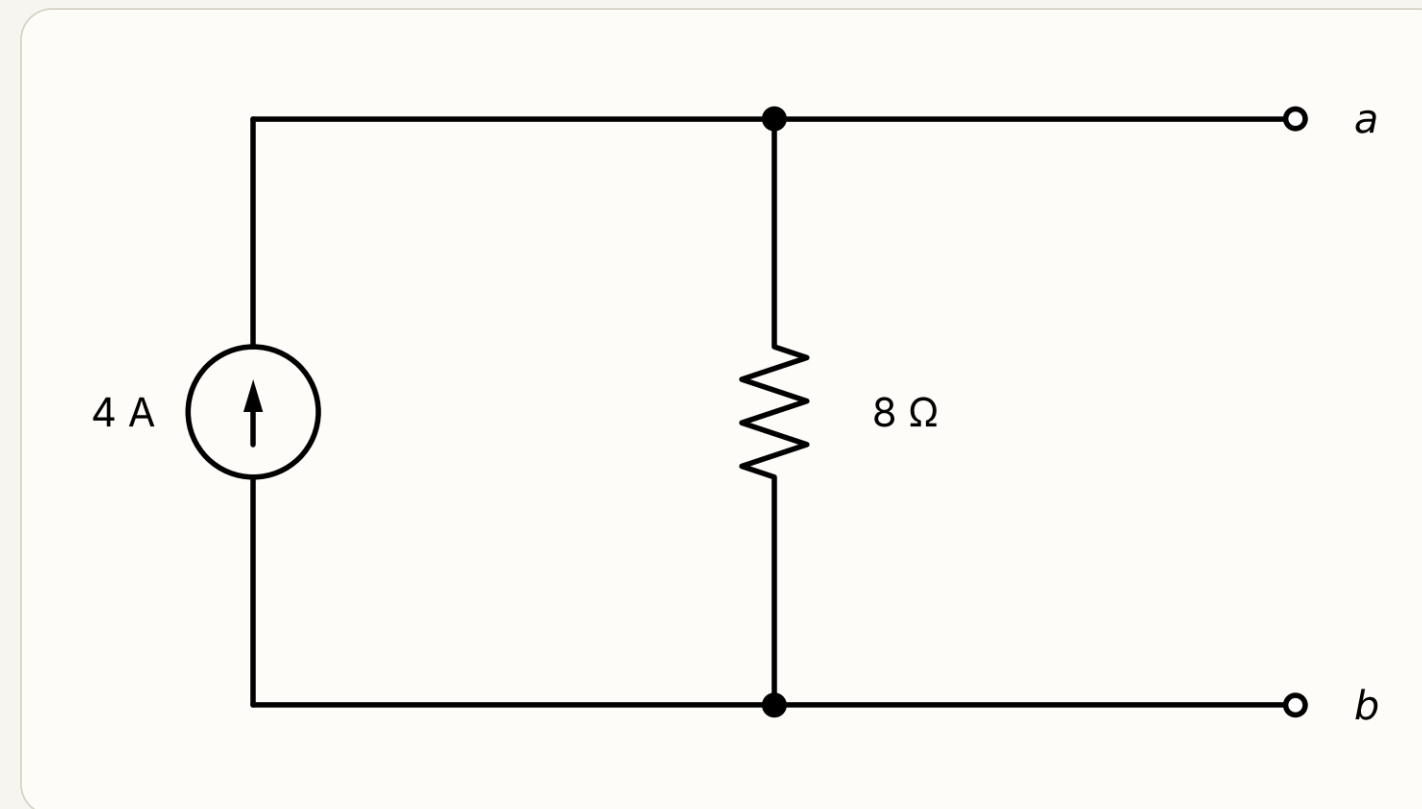
THÉVENIN AND NORTON EQUIVALENTS

Any linear circuit is one source and one resistor

Thévenin



Norton

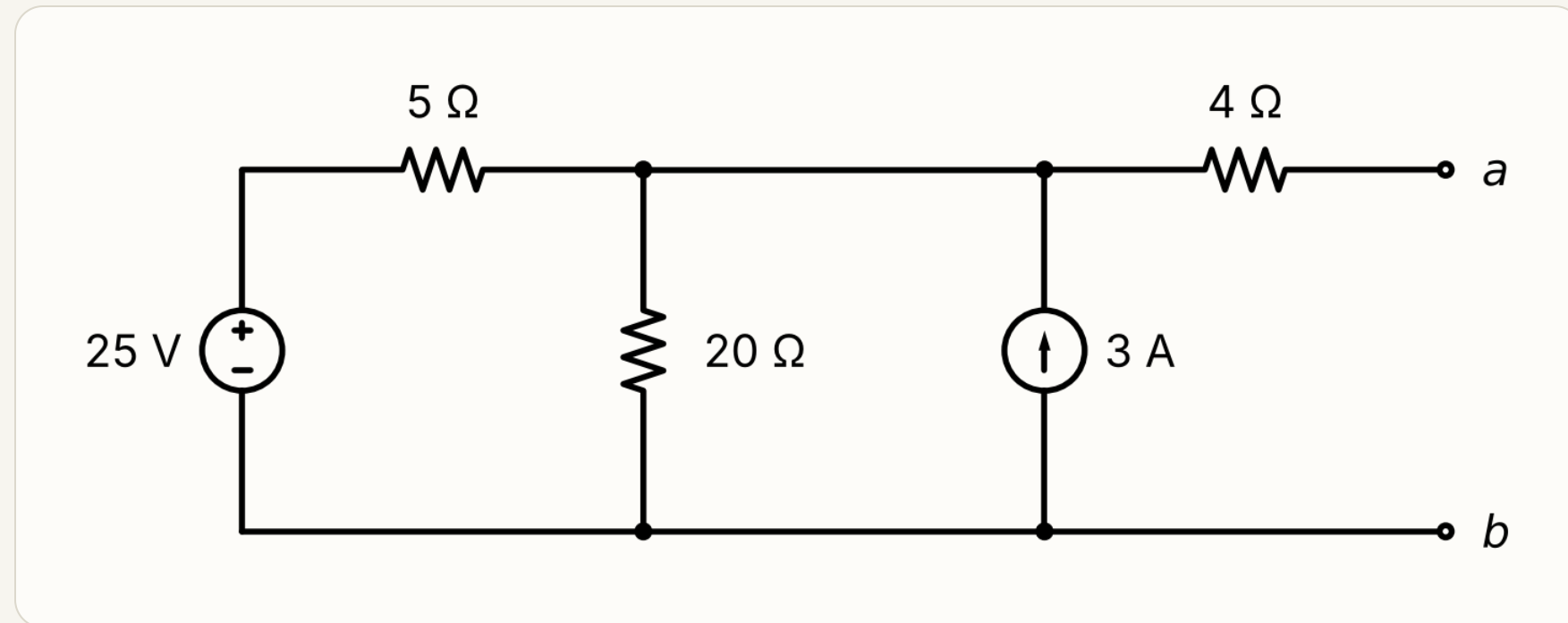


$$V_{Th} = v_{oc}, \quad R_{Th} = \frac{v_{oc}}{i_{sc}}, \quad I_N = i_{sc}$$

Seen from terminals a and b. Find any two of v_{oc} , i_{sc} and R_{Th} ; the third follows. A source transformation turns one equivalent into the other.

NILSSON EXAMPLE 4.14

V_{Th} is the open-circuit voltage



With a and b open, the 4 Ω carries no current, so $v_{ab} = v_1$:

$$\frac{v_1 - 25}{5} + \frac{v_1}{20} - 3 = 0 \Rightarrow v_1 = 32 \text{ V} = V_{Th}$$

NILSSON EXAMPLE 4.14

Short a to b for i_{sc} , then $R_{Th} = V_{Th}/i_{sc}$

With a and b shorted, the 4 Ω carries the short-circuit current:

$$\frac{v_2 - 25}{5} + \frac{v_2}{20} - 3 + \frac{v_2}{4} = 0 \Rightarrow v_2 = 16 \text{ V}$$

$$i_{sc} = \frac{16}{4} = 4 \text{ A}, \quad R_{Th} = \frac{32}{4} = 8 \text{ } \Omega$$

Check: a 24 Ω load gets 24 V and 1 A, from the original circuit and from 32 V behind 8 Ω alike.

NILSSON EXAMPLE 4.15

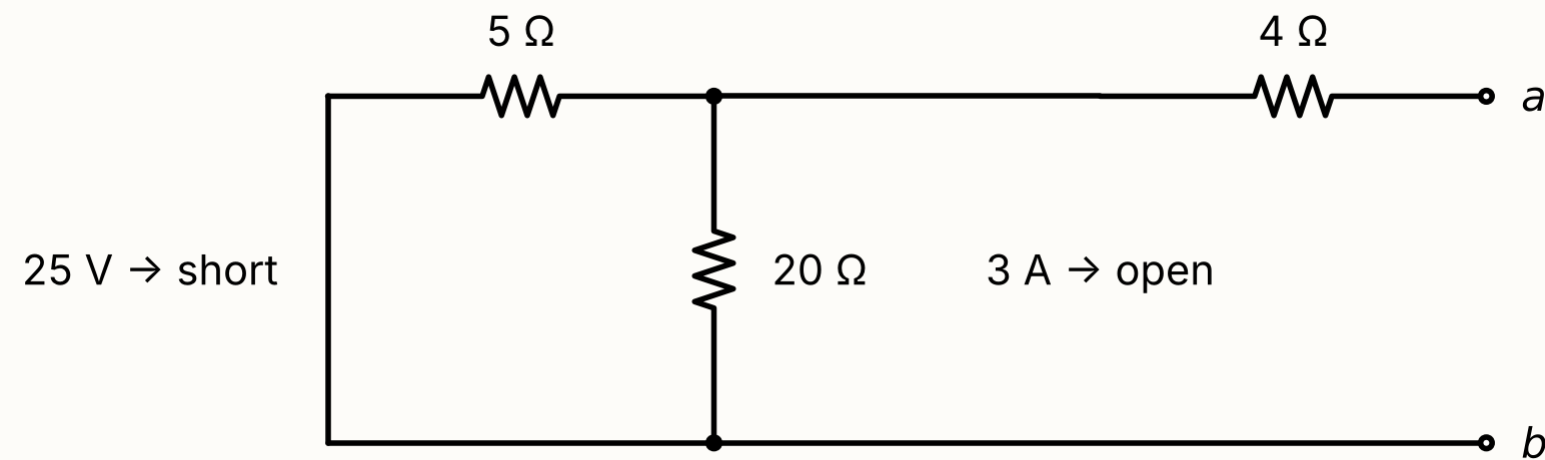
Source transformations give the Norton equivalent directly

- 1 25 V and 5 Ω become 5 A in parallel with 5 Ω .
- 2 5 A + 3 A = 8 A, both pointing up; 5 $\parallel$ 20 = 4 Ω .
- 3 8 A with 4 Ω becomes 32 V in series with 4 Ω ; add the 4 Ω : 32 V behind 8 Ω (Thévenin).
- 4 32 V and 8 Ω become 4 A in parallel with 8 Ω (Norton).

Dependent sources can be transformed too, but never transform away the element that carries a controlling voltage or current.

NILSSON EXAMPLE 4.17

R_{Th} directly: deactivate the independent sources



The Example 4.14 circuit, sources deactivated:

$$R_{Th} = 4 + \frac{(5)(20)}{5 + 20} = 8 \, \Omega$$

Voltage source

Becomes a short circuit: its voltage is 0.

Current source

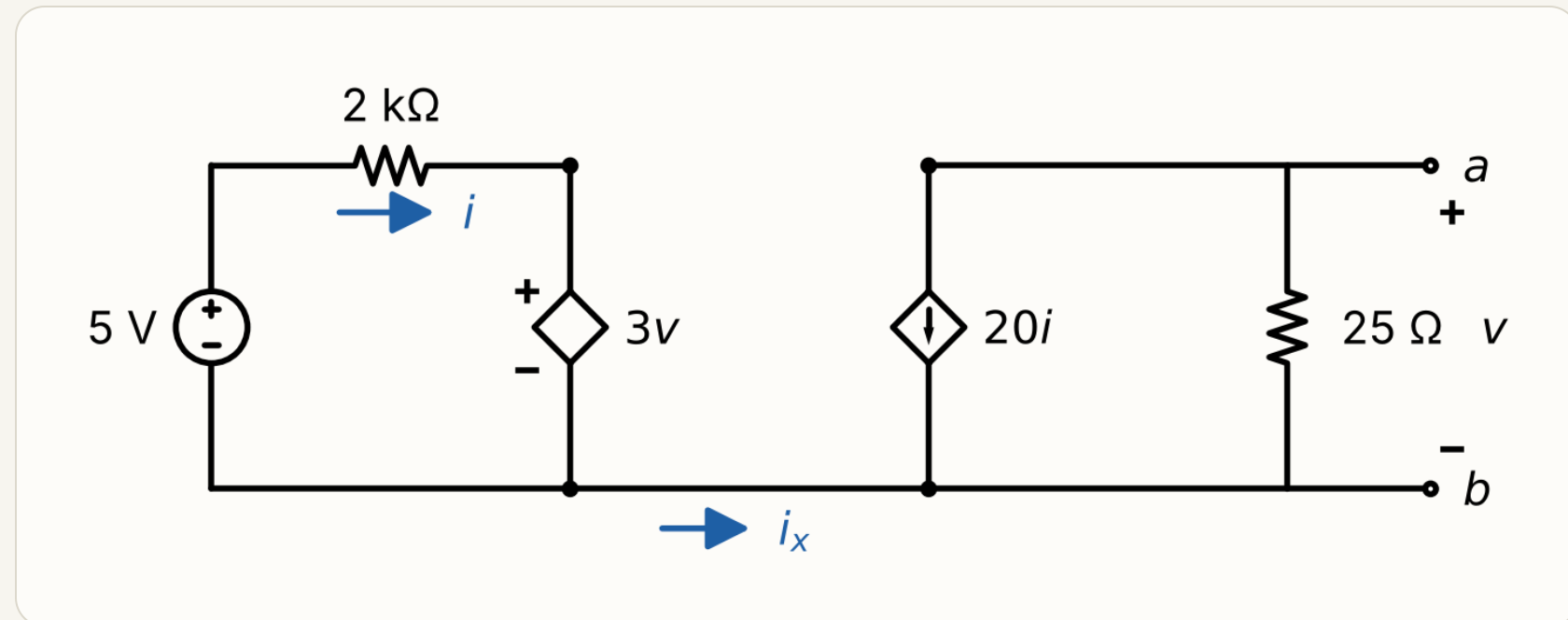
Becomes an open circuit: its current is 0.

Dependent source

Stays. Use V_{Th}/i_{sc} or a test source (slides 16–17).

NILSSON EXAMPLE 4.16

Dependent source: find V_{Th} and i_{sc} , never deactivate it



$i_x = 0$: there is no return path between the two halves.

v is v_{ab} , so it is zero when a and b are shorted.

$$V_{Th} = -20i(25), \quad i = \frac{5 - 3V_{Th}}{2000} \Rightarrow V_{Th} = -5 \text{ V}$$

$$v = 0: i = \frac{5}{2000} = 2.5 \text{ mA}, \quad i_{sc} = -20i = -50 \text{ mA}, \quad R_{Th} = \frac{-5}{-0.05} = 100 \text{ } \Omega$$

NILSSON EXAMPLE 4.18

R_{Th} with a test source

Deactivate the 5 V source only. Apply v_T at a–b and find the current i_T it delivers:

$$i_T = \frac{v_T}{25} + 20i, \quad i = \frac{-3v_T}{2000} \Rightarrow \frac{i_T}{v_T} = \frac{1}{25} - \frac{3}{100} = \frac{1}{100}$$

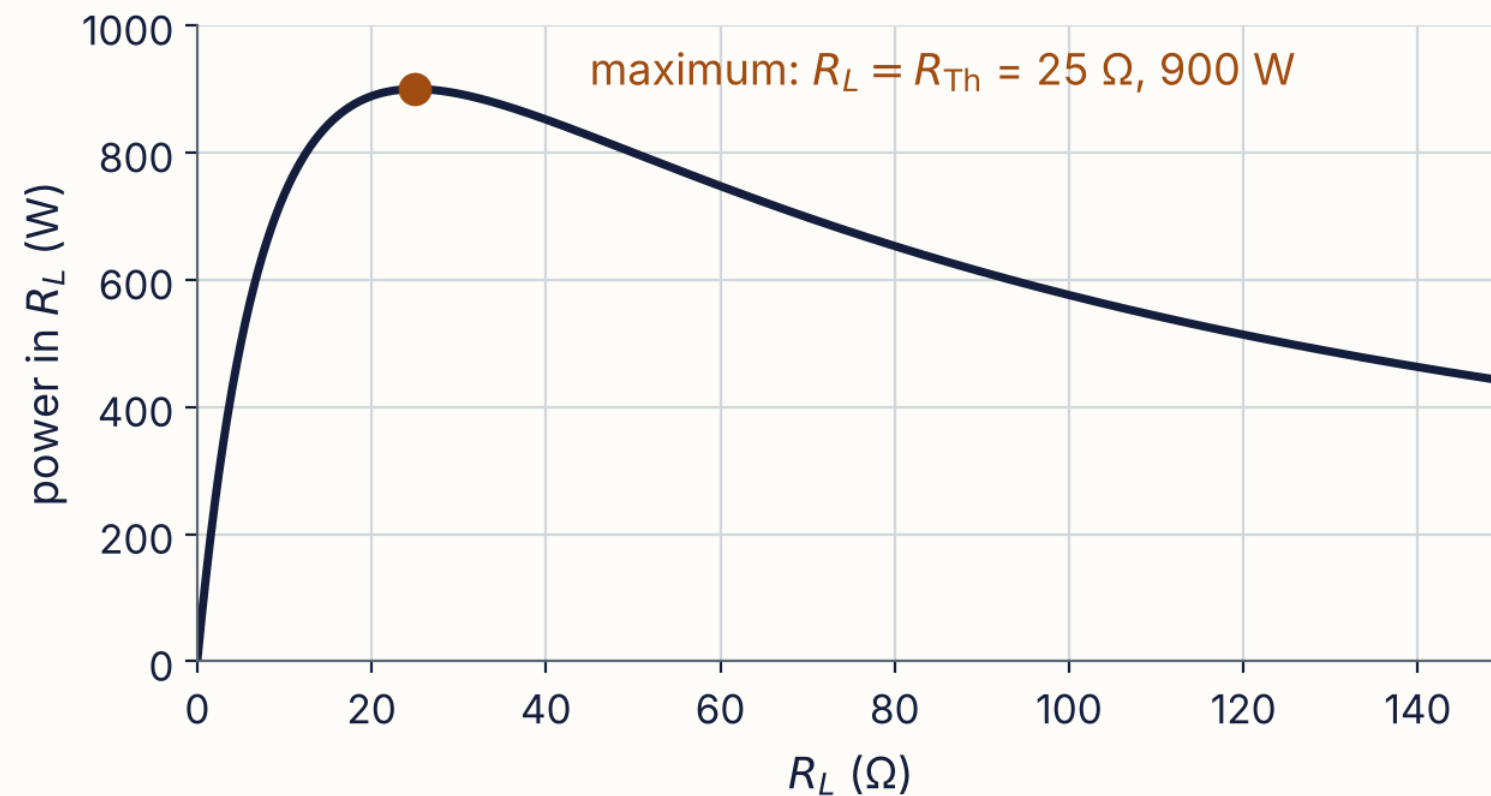
$R_{Th} = v_T/i_T = 100 \, \Omega$, the same as V_{Th}/i_{sc} .

Only dependent sources?

Then $V_{Th} = 0$ and $I_N = 0$, and the equivalent is one resistor, found only with a test source (Nilsson Example 4.19: $90 \, \Omega$).

MAXIMUM POWER TRANSFER

A load gets the most power when $R_L = R_{Th}$



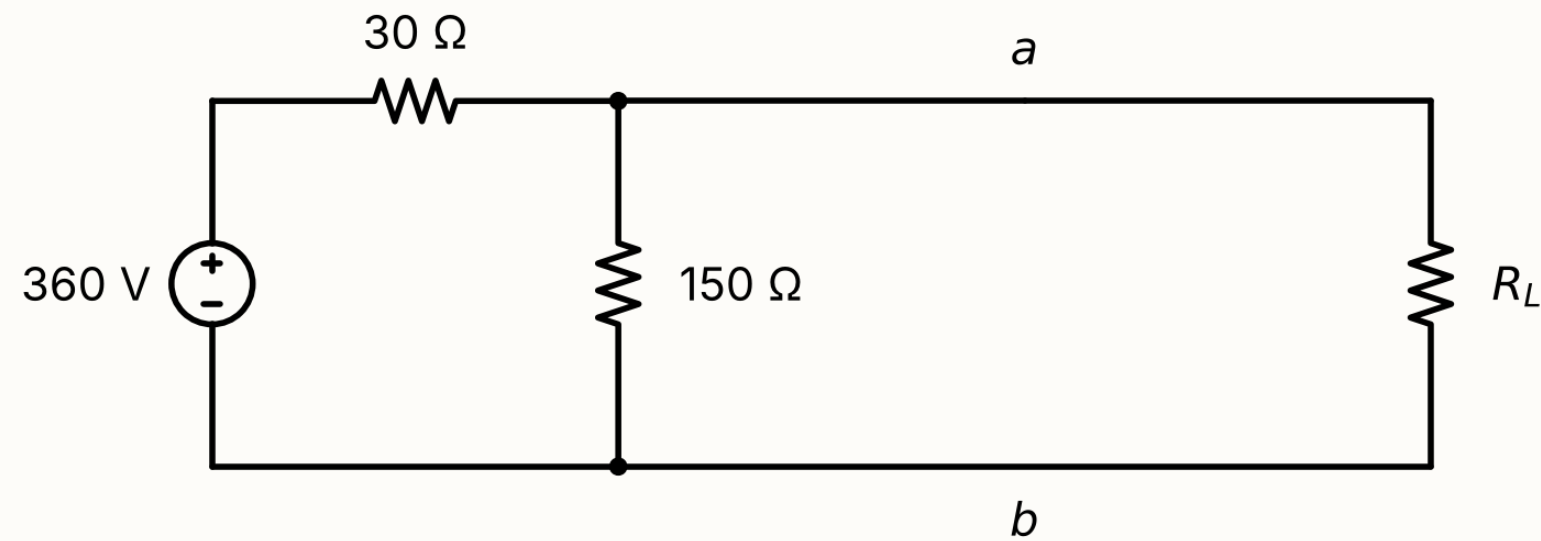
$$p = \frac{V_{Th}^2 R_L}{(R_{Th} + R_L)^2}$$

$$\text{largest at } R_L = R_{Th}: \quad p_{\max} = \frac{V_{Th}^2}{4R_{Th}}$$

Curve: Example 4.21, with 300 V behind 25 Ω .

NILSSON EXAMPLE 4.21

Find R_L , $p_{\max}$ and the share that arrives



$$V_{Th} = \frac{150}{180}(360) = 300 \text{ V}, \quad R_{Th} = \frac{(30)(150)}{180} = 25 \text{ } \Omega$$

With $R_L = 25 \text{ } \Omega$, $v_{ab} = 150 \text{ V}$ and the 360 V source delivers 2520 W :

$$p_{\max} = \frac{300^2}{(4)(25)} = 900 \text{ W}, \quad i_s = \frac{360 - 150}{30} = 7 \text{ A}, \quad \frac{900}{2520} = 35.7\%$$

MAXIMUM POWER IS NOT MAXIMUM EFFICIENCY

Who wants it, and who avoids it

At maximum power

The load gets half of the Thévenin source's power. In Example 4.21 only 35.7% of the 360 V source's power arrives.

Power systems

They keep R_L far above R_{Th} : little is lost and the voltage holds. The generator on slide 2 is one (20 Ω heater, 3 Ω behind it).

Signal circuits

Antennas and sensors have little power to spare, so matching the load is worth the loss.

CHOOSING A TOOL

Each tool fits a kind of question

Tool	Use it when	Watch out for
Source transformation	Series V–R and parallel I–R branches reduce step by step	The arrow points toward the + side; keep the branch you need
Superposition	You want one source's effect, or the sources differ in kind	Short an independent voltage source, open an independent current source; add currents, not powers
Thévenin or Norton	One load changes, or maximum power is asked	Never deactivate a dependent source; check the short-circuit current
Node or mesh (Part 1)	Everything else, and every check	—

CHECKING AN AI ANSWER

An AI transformed Example 4.15's 25 V source like this. What is wrong?

$$25 \text{ V}, 5 \Omega \rightarrow 5 \text{ A pointing down: } 3 - 5 = -2 \text{ A}, \quad V_{\text{Th}} = -8 \text{ V}, \quad R_{\text{Th}} = 8 \Omega$$

Correct:

$$5 \text{ A pointing up: } 5 + 3 = 8 \text{ A}, \quad V_{\text{Th}} = (8)(4) = 32 \text{ V}$$

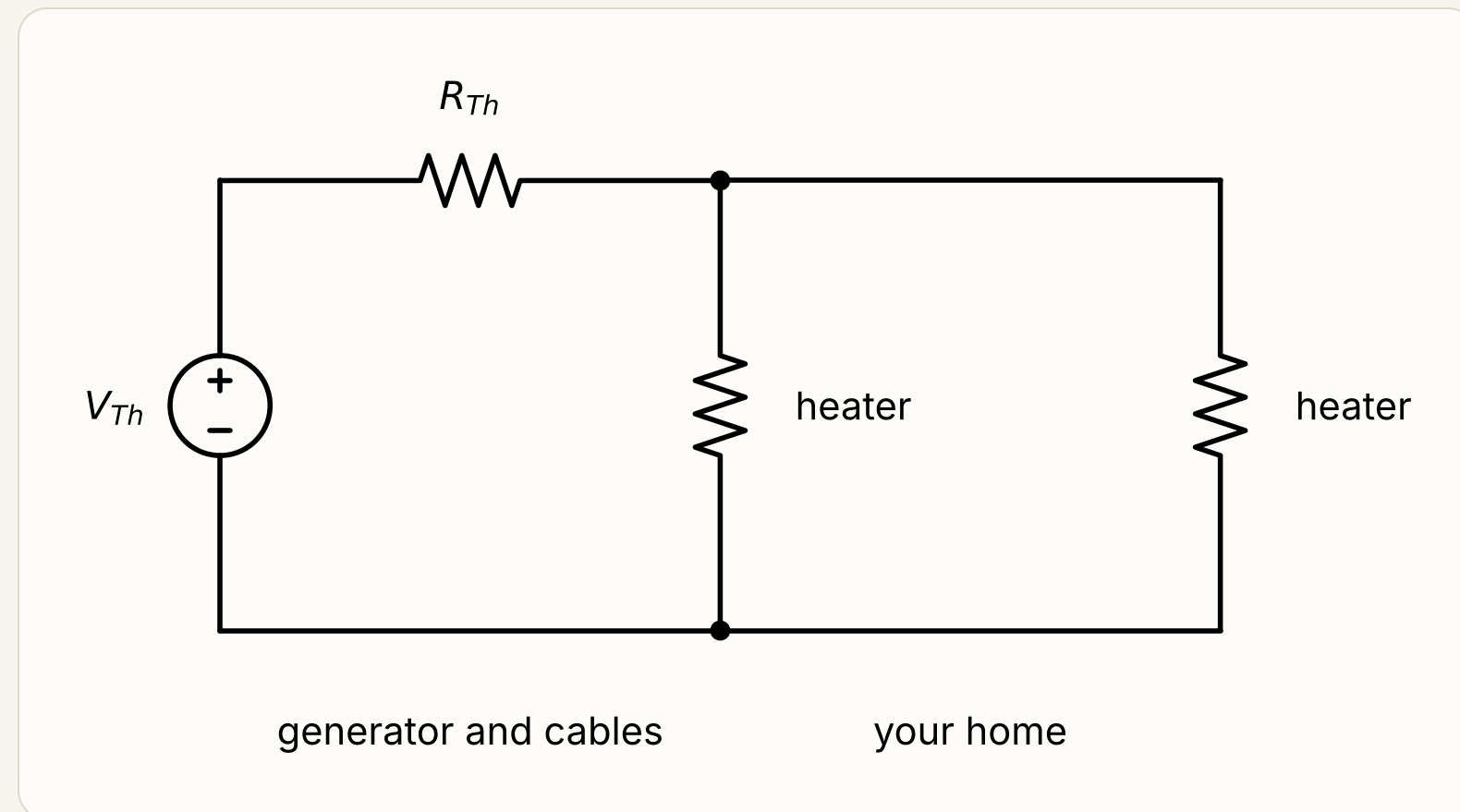
The arrow points toward the + terminal. The giveaway: KCL at the top node fails with the AI's V_{Th} ; its R_{Th} of 8Ω is right and cannot catch the slip.

$$\frac{-8 - 25}{5} + \frac{-8}{20} - 3 = -10 \text{ A} \neq 0$$

In a 2025 study of Gemini 2.5 Pro on undergraduate circuit problems, misread source polarities caused about a third of its wrong answers, and misread current directions nearly as many.

BACK TO THE GENERATOR

Two readings give the Thévenin equivalent



One heater: 10 A at 200 V.

$$R_{Th} = \frac{230 - 200}{10} = 3 \, \Omega, \quad R_{heater} = \frac{200}{10} = 20 \, \Omega$$

Two heaters, 10 Ω :

$$v = 230 \frac{10}{10 + 3} = 176.9 \, \text{V}$$

$$p = \frac{176.9^2}{20} = 1565 \, \text{W per heater (2000 W with one heater on)}$$

A thicker cable lowers R_{Th} and the drop.

WHAT COMES NEXT

Practise openly, then show it on your own

In class

Booklet Problems 4.13 to 4.27.
Problem 4.28 is homework.

At home

The 16-item practice sheet. AI
tools allowed; not marked.

Week 10

15-minute quiz, no devices.
Every question is a twin of a
practice-sheet question.

Chapter 5 (from week 9, session 2): operational amplifiers, whose model uses a resistance at the input and a Thévenin equivalent at the output.

CHECKLIST

You can do these without help

- Transform a voltage source with a series resistor into a current source with a parallel resistor, with the arrow the right way, and back
- Use superposition to find a current, and explain why powers cannot be added
- Find V_{Th} , i_{sc} and R_{Th} , and draw the Thévenin and Norton equivalents
- Find R_{Th} by deactivating the independent sources, and with a test source when the circuit has a dependent source
- Choose the load for maximum power transfer and find that power and the efficiency
- Check an equivalent with its short-circuit current, $i_{sc} = V_{Th}/R_{Th}$